MALAWI NATIONAL EXAMINATIONS BOARD

2024 MALAWI SCHOOL CERTIFICATE OF EDUCATION (MSCE) EXAMINATION

MATHEMATICS - PAPER I

Subject Code: M071/I | Time Allowed: 2 Hours

INSTRUCTIONS TO CANDIDATES:

1. This paper contains two sections: Section A and Section B.
2. Section A contains 10 compulsory short-answer questions (55 Marks).
3. Section B contains 5 extended-response questions; answer any THREE (45 Marks).
4. Mathematical tables and non-programmable electronic calculators may be used.
5. All working must be clearly shown in the spaces provided.
6. Silent electronic calculators may be used where necessary.

CANDIDATE IDENTIFICATION:

Candidate Examination Number: [_____]
Examination Centre Name: [_____]
Date of Examination: [October 2024]

Official Ministry of Education National Secondary Curriculum Document.
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SECTION A (55 MARKS)

Answer ALL questions in this section.

Question 1: Algebra & Factorization

(a) Simplify completely: (3x^2 - 12) / (x^2 + 5x + 6) [3 marks]

(b) Solve the quadratic equation: 2x^2 - 7x + 3 = 0 [3 marks]

Question 2: Logarithms & Indices

(a) Given that log_10(2) = 0.3010 and log_10(3) = 0.4771, evaluate log_10(72). [3 marks]

(b) Solve for x: 9^(x-1) = 27^(x+2) [3 marks]

Question 3: Coordinate Geometry & Vectors

Points P(2, -3) and Q(6, 5) lie on a Cartesian plane.

(a) Calculate the length of the line segment PQ. [2 marks]

(b) Find the equation of the perpendicular bisector of PQ. [4 marks]

Question 4: Commercial Mathematics & Matrices

A trader buys 5 bags of maize and 3 bags of beans for MWK 145,000.

Express this as a matrix equation and determine the cost per bag. [5 marks]

SECTION B (45 MARKS)

Answer any THREE questions from this section.

Question 5: Functions, Graphs & Calculus

A function is defined by f(x) = 2x^3 - 3x^2 - 12x + 8.

(a) Find the derivative f’(x) and determine the stationary points. [5 marks]

(b) Distinguish between the local maximum and minimum coordinates. [4 marks]

(c) Sketch the curve f(x) for the interval -2 <= x <= 3. [6 marks]

Question 6: Statistics & Probability

The table below represents the test scores of 60 Form 4 candidates:

Score Interval: 40-49 | 50-59 | 60-69 | 70-79 | 80-89 | 90-99

Frequency: 6 | 12 | 18 | 14 | 7 | 3

(a) Calculate the estimated mean score using an assumed mean of 64.5. [6 marks]

(b) Draw a cumulative frequency curve (ogive) and estimate the median. [6 marks]

(c) Determine the interquartile range (IQR). [3 marks]